

High Availability Architecture — Study Notes

Active-Passive · Active-Active · Multi-Master DB
Failover · Replication · Single Point of Failure · 99.999%

TOPICS COVERED

1. What is High Availability?

2. Single Node — The Problem

3. Active-Passive Architecture

4. Active-Passive Trade-offs

5. Active-Active — Deep Dive

6. Synchronisation & Conflicts

7. Architecture Comparison

8. Quick Revision & Interview Qs

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1. What is High Availability?

High availability (HA) is a system design property that ensures a service remains operational and accessible for the maximum possible time. The industry standard target is **five nines — 99.999% uptime**, which translates to less than 5.26 minutes of downtime per year.

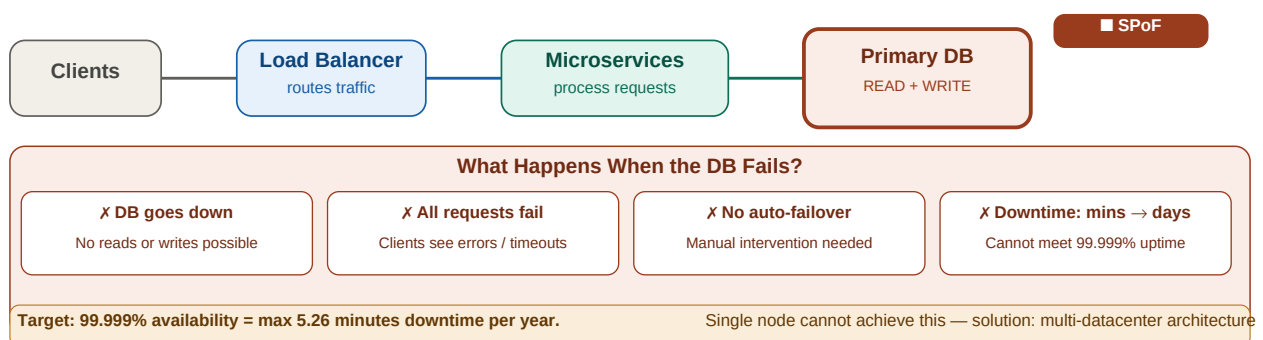
Availability	Downtime/Year	Downtime/Month	Suitable for
99%	3.65 days	7.2 hours	Development, internal tools
99.9%	8.76 hours	43.8 minutes	Small business apps
99.99%	52.6 minutes	4.4 minutes	E-commerce, SaaS
99.999% (5 9s)	5.26 minutes	26 seconds	Banking, payments, critical infra

Goal	What it means	How to achieve
High Availability	Service stays up 99.999% of the time	Eliminate SPoF, add redundancy
Data Resiliency	System recovers after failure without data loss	Replication across multiple nodes
Fault Tolerance	System continues despite component failures	Active-active or active-passive

2. Single Node Architecture — The Problem

The simplest architecture uses a single primary database. It is easy to build and cheap — but has a critical flaw: the database is a **Single Point of Failure (SPoF)**. When it goes down, everything goes down.

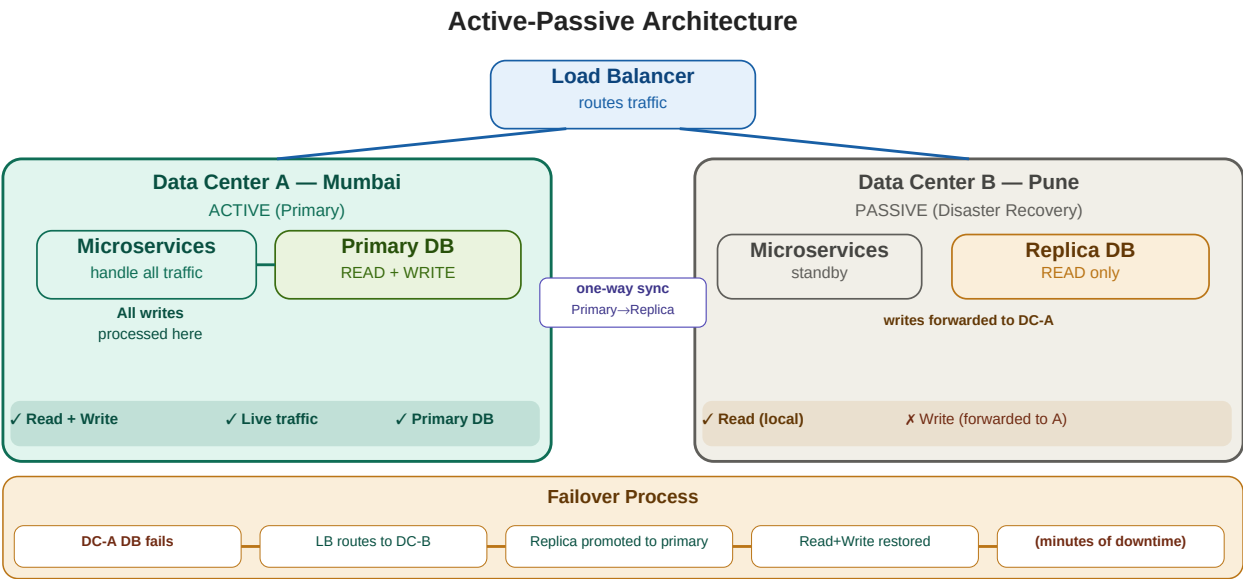
Single Node Architecture — The Problem



Single node: DB failure = total system failure · No auto-recovery · Cannot achieve 99.999% uptime

3 & 4. Active-Passive Architecture

Active-passive architecture uses two data centers. One is the **active (primary)** handling all read/write traffic. The other is the **passive (disaster recovery)** with a replica database that stays in sync via one-way replication. On failure, the replica is promoted to primary.



Active-Passive: DC-A (Mumbai) active, DC-B (Pune) standby · One-way replication · Writes forwarded to primary · Failover on failure

Aspect	Active DC	Passive DC
Handles reads	Yes — from local DB	Yes — from replica (read replica)
Handles writes	Yes — directly to primary DB	Forwarded to Active DC (extra latency)
DB type	Primary (read-write)	Replica (read-only by default)
On failure	Traffic rerouted to Passive	Replica promoted to Primary
Resource usage	100% active	Idle until failover

Advantages

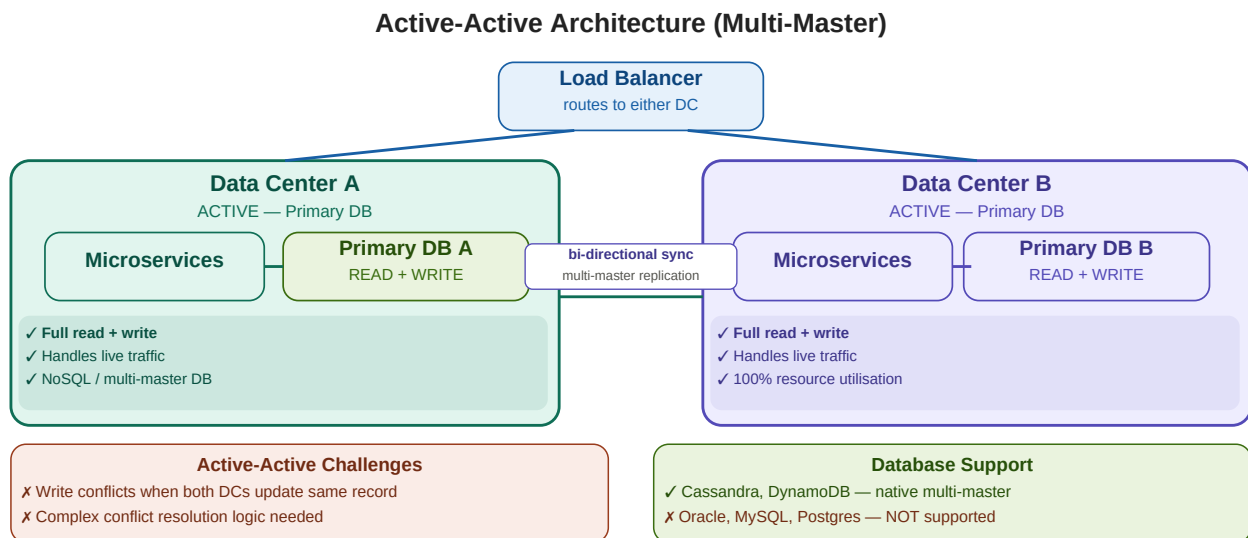
- Eliminates single point of failure — one DC down, the other takes over
- Provides data resiliency — replica has a full copy of data
- Supports read scaling — passive DC can serve read requests locally
- Familiar, well-supported by traditional RDBMS (Oracle, MySQL, Postgres)

Disadvantages

- Resource underutilisation — passive DC's write capacity sits idle
- Write latency for passive DC users — all writes forwarded to active DC
- Failover delay — promoting replica takes minutes, causing brief downtime
- Not suitable for write-heavy apps — all writes bottleneck on one primary
- Failover is NOT automatic in basic setups — requires monitoring + intervention

5 & 6. Active-Active Architecture (Multi-Master)

Active-active architecture has **both data centers fully operational** simultaneously. Each has its own primary (read-write) database. Traffic is load balanced across both. Data is synchronised bidirectionally — a write to DC-A is replicated to DC-B and vice versa.



Active-Active: both DCs handle reads+writes · Bidirectional multi-master sync · 100% resource utilisation · Complex conflict resolution

Advantages

- 100% resource utilisation — both DCs handle traffic simultaneously
- Higher write and read throughput — load distributed across two primaries
- Seamless failover — if one DC fails, the other already handles full traffic
- Better geographic distribution — users routed to nearest data centre

Challenges

- Write conflicts: both DCs can update the same record simultaneously
- Complex conflict resolution: last-write-wins, application-level merge, or versioning
- Eventual consistency: read from DC-B may not reflect latest write to DC-A immediately
- Not supported by traditional RDBMS — Oracle, MySQL, Postgres have single-master constraint
- Requires purpose-built multi-master databases

Cassandra: native multi-master replication. Each node accepts writes. Conflict resolution via timestamp (last-write-wins).

DynamoDB: global tables support multi-master across regions. Uses version-based conflict resolution.

MySQL/Postgres: support read replicas (active-passive) but NOT multi-master writes natively.

CockroachDB / YugabyteDB: distributed SQL databases that support multi-region writes.

7. Architecture Comparison

Architecture Comparison			
Feature	Single Node	Active-Passive	Active-Active
High Availability	No	Yes	Yes
Data Resiliency	No	Yes	Yes
Single Point of Failure	Yes	No	No
Write Scalability	Low	Low (one primary)	High (multi-master)
Read Scalability	Low	Moderate (replicas)	High
Failover Time	Total failure	Minutes (promotion)	Instant (seamless)
Resource Usage	N/A	Poor (passive idle)	Optimal (100%)
Setup Complexity	Low	Moderate	High
DB Compatibility	Any	Any RDBMS	NoSQL / multi-master
Sync Complexity	None	One-way, simple	Bi-dir + conflict res.

Single Node vs Active-Passive vs Active-Active across 10 key dimensions

8. Quick Revision & Interview Questions

Quick Revision

- > High availability = system stays up 99.999% of the time = max 5.26 min downtime/year.
- > Single point of failure (SPoF): one component whose failure causes total system failure.
- > Single node DB is a SPoF — when it fails, all reads and writes fail.
- > Active-passive: one active DC (read+write), one passive DC (replica, read-only).
- > Active-passive replication is ONE-WAY: primary → replica.
- > Writes to passive DC are FORWARDED to the active DC — adds latency.
- > Failover in active-passive: replica is promoted to primary — takes minutes.
- > Passive DC is underutilised — its write capacity sits idle during normal operation.
- > Active-active: BOTH DCs fully active, each with its own primary DB (multi-master).
- > Active-active replication is BI-DIRECTIONAL — conflicts can arise.
- > Conflict resolution needed when two DCs write to the same record simultaneously.
- > Active-active: 100% resource utilisation, seamless failover, higher throughput.
- > Traditional RDBMS (Oracle, MySQL, Postgres) do NOT support multi-master writes.
- > NoSQL (Cassandra, DynamoDB) natively support multi-master — suited for active-active.
- > CAP theorem applies: active-active sacrifices consistency for availability + partition tolerance.

Interview Questions

1. What is high availability? What does 99.999% uptime mean in practice?
2. Why is a single node database a problem for high availability?
3. Explain active-passive architecture. How does failover work?
4. What is the disadvantage of forwarding writes to the active DC in active-passive?
5. What is active-active architecture and how does it differ from active-passive?
6. What is a multi-master database? Give examples.

- 7. Why can't MySQL be used easily in an active-active architecture?
- 8. What is a write conflict in active-active? How is it resolved?
- 9. What is eventual consistency and when does it occur in active-active?
- 10. Compare active-passive and active-active on: failover time, resource usage, complexity.
- 11. When would you choose active-passive over active-active?
- 12. Design a high-availability database architecture for a payment system.

Key Terms

High Availability	99.999% uptime — max 5.26 minutes downtime per year
SPoF	Single Point of Failure — one component whose failure breaks everything
Data Resiliency	Ability to recover and continue functioning after a failure
Active-Passive	One active DC + one standby. Replica promoted on failure.
Active-Active	Both DCs fully active with multi-master DBs and bidirectional sync
Multi-Master DB	Database where multiple nodes accept write operations simultaneously
Replication	Copying data from one DB node to another to keep them in sync
One-way Replication	Primary → Replica only. Used in active-passive.
Bi-directional Sync	Both nodes replicate to each other. Used in active-active.
Failover	Switching traffic to standby when primary fails
Write Conflict	Two DCs update same record simultaneously — needs resolution
Eventual Consistency	Data converges to same state over time — may be briefly stale
Five Nines (99.999%)	Industry standard for HA — 5.26 minutes max downtime/year
Disaster Recovery	Passive DC exists solely to recover from catastrophic failure